

Weekly Lesson Plan – Week 3 • Semester 1 • 2026–27 • Da Vinci School

Teacher: Gavaldon Week of: Aug 24, 2026 Course: Engineering Design & Problem Solving (InSPIRESS)				
TEKS / ELPS:				
TEKS: §127.406(12) develop and present a comprehensive report describing the problem and solution; §127.406(9) use structured techniques to select and justify a preferred solution; §127.406(8) conduct research and analyze data to create a clearly defined problem statement; §127.406(9) use structured techniques to select and justify a preferred solution; (12) present the reasoning; §127.406(6) implement a design process to develop a multi-system product or solution ELPS: c.1A, c.2E, c.3D, c.3E, c.4G, c.5B				
Aim of the Lesson / Objective: <i>Statement of what you want students to do (skill) and/or know (knowledge). Students will...</i>				
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
Students will present their product to a review panel and defend it under questioning, and evaluate peer designs against a shared rubric.	Students will finish presenting and defending their designs, then identify what the strongest design solutions had in common.	Students will analyze an engineering design failure and state the problem, the stakeholders, and what evidence is missing.	Students will generate three design responses, weigh the trade-offs, and defend a decision with evidence.	Students will follow a guided build to produce a working program and debug it using the error messages.
Employed Language Domain(s) Objective: <i>How students achieve the objective – Listening, Speaking, Reading, Writing. Students will...</i>				
Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.				
Assessment of Content Objective: <i>Instrument used to determine mastery. Ex: Exit Ticket.</i>				
Students will be assessed by written, group, or oral responses throughout the class period.				
Set (Bell Work): <i>A motivating action that engages students in the objective – question, fact, activity. Essential Question:</i>				
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
What is the one question you are most afraid the panel will ask you?	Which product you saw yesterday solved the most real problem, and why?	Your design passes every test you ran, and then fails for a real user. What does that tell you about your testing?	Is a design decision still good if it turns out badly? Explain.	What is the difference between let and const?
Methodology / Steps: <i>What it takes to teach the objective so 100% can pass the assessment. Teacher will... Teams will... Individuals will...</i>				
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
Step 1: Bell ringer / essential question. Step 2: Bell-work: the question you're most afraid of (5 min) Step 3: Order of business + judging rules (5 min) Step 4: Design Reviews: 3-min pitch + 2-min defense per firm (30 min) Step 5: Turn in peer scorecards (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: which product solved the most real problem? (5 min) Step 3: Remaining firms present + defend (28 min) Step 4: Class debrief: what every strong pitch had (7 min) Step 5: Best Engineering Judgment + exit (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell work (priming question) - 10 minutes, written (10 min) Step 3: Assign scenarios + explain the four steps (5 min) Step 4: Work the scenario: problem, who is affected, what you still need to know, the real pressure (23 min) Step 5: Table talk: read your one-sentence problem to your neighbor (7 min)	Step 1: Bell ringer / essential question. Step 2: Bell work (priming question) - 10 minutes, written (10 min) Step 3: What a real reason sounds like (4 min) Step 4: Three options, what each costs, your call, three reasons (19 min) Step 5: 60-second report outs + one question each (12 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: let versus const (5 min) Step 3: How a Workshop works + the stuck-protocol (5 min) Step 4: Build a Greeting Bot (30 min) Step 5: Exit ticket: paste the line that prints the message (5 min)
Resources: <i>Textbook, handouts, visuals, laptops.</i>				
Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.				
Re-Teaching Activities: <i>What you prepare if students do not master the objective.</i>				
Review during bellwork; small-group re-teach; peer support.				
Reflection: <i>Students analyze what they accomplished and how they reached mastery.</i>				
Exit ticket – students state what they accomplished and what worked best.				
Study Skills: <i>Skills implemented by students – note-taking, summarizing, teamwork, etc.</i>				
Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.				
SPED & 504 Accommodations / Modifications: <i>Codes.</i>				
Per Individualized Education Plan (IEP) / 504 plan.				
ELL Accommodations: <i>Codes.</i>				
Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP).				